

# Chapter 24 - Memory Model and MMIO Map

Part IV of this guide is the bridge between BASIC and machine language. From here on, every chapter assumes you want to read and write the Intuition Engine's hardware directly: from BASIC with PEEK and POKE, from the machine monitor, or from a program written in one of the six processor languages described in Chapters 26 through 30.

Everything you do at that level happens through one shared physical bus. This chapter is the map of the low fixed regions of that bus, and the rules for CPUs that see narrower address spaces.

## 24.1 The address space

Intuition Engine has one 64-bit physical bus. It carries 64-bit physical addresses and accepts 8, 16, 32, and 64-bit transfers. The classic chips and helper devices are not separate computers; they are devices and adapters on that bus.

The low 4 gigabytes, \$00000000 to \$FFFFFFFF, form the legacy CPU and device window used by BASIC examples, MMIO registers, video apertures, and the compatibility processors. A MOVE.L D0, (\$F0700).L from M68K, a STORE.L (R5), R6 from IE64 with R5 = \$F0700, and a BASIC POKE &H000F0700,n all arrive at the same byte in that low window.

IE64 can address the full 64-bit physical range. Its MMU uses 64-bit virtual addresses and page-table entries with a 52-bit physical page number field, giving 64-bit physical addresses after the 4 KB page offset is added. Above the legacy low-memory slice, RAM is supplied by the machine's guest-RAM backing and discovered through SysInfo and CR\_RAM\_SIZE\_BYTES.

This is not how every processor sees the world. IE32, M68K, and x86 have 32-bit address registers and see the low 4 GB window directly. The 6502 and Z80 can address only 64 kilobytes directly. Each of those CPUs has a small set of apertures into the larger bus; the apertures are described in the per-CPU chapters.

### 24.1.1 Three zones

The fixed low window divides into three zones:

| Range                   | Size    | Contents                                     |
|-------------------------|---------|----------------------------------------------|
| \$00000000 - \$0009EFFF | 636K    | Main RAM (low memory, code and data)         |
| \$0009F000 - \$000FFFFF | 388K    | Stack, VRAM apertures and MMIO               |
| \$00100000 - \$005FFFFF | 5M      | Main VRAM                                    |
| \$00600000 - \$FFFEFFFF | Dynamic | Extended RAM (sized at boot)                 |
| \$FFFF0000 - \$FFFFFFFF | 64K     | Sign-extended alias of \$00000000-\$0000FFFF |

The 5-megabyte VRAM block from \$100000 to \$5FFFFFF is wired directly to the VideoChip framebuffer (Chapter 4). Writes here appear on screen the next time the compositor refreshes.

Memory beyond \$600000 is plain RAM when it is backed by the current guest-RAM allocation. Its upper bound depends on the amount of RAM the Intuition Engine was started with and on the active CPU profile. The two words at \$F2400/\$F2404 (SYSINFO\_TOTAL\_RAM\_L0/HI) report the byte size of total guest RAM as a 64-bit value; \$F2408/\$F240C

report the size visible to the currently executing CPU profile. \$F2414/\$F2418 report the dense low RAM window: the contiguous low-memory span that 32-bit clients, flat images, and low-window device buffers can address directly.

IE64 BASIC keeps its own programme text, variables, stacks, file bridges, line/input scratch, and compiler scratch areas in dynamic reservations inside the same address space. In the normal low32 fallback layout the line/input scratch begins at \$01000000, and the programme, variable, string, and file-bridge arena begins after that scratch reservation. The current pointer and capacity are BASIC state, not a fixed public line-buffer address.

Reader programmes should not depend on those private reservations. When BASIC needs a public buffer for a copper list, coprocessor request, file staging area, or DMA-style hardware block, use `MEMALLOC(size[, align])`. It allocates from public low32 ranges that are meant to be shared with devices and other CPUs.

The low memory slice and the backed RAM above it are both active RAM, but the boundary between them matters for scalar accesses. Byte reads and writes work on either side. Word and long reads and writes must fit wholly inside the low slice or wholly inside backed RAM. If a word or long starts before the boundary and ends after it, the access is unmapped: a read returns zero and a write does not partly update either side.

Byte-copy devices can still cross the same boundary one byte at a time. The File I/O block in Chapter 35 uses that rule when it copies a file into a destination buffer. Device MMIO is still MMIO, not RAM, even when its address lies inside the low 32-bit window.

### 24.1.2 The sign-extended alias

The top 64 kilobytes (\$FFFF0000 - \$FFFFFFFF) are not real memory of their own. They are a mirror of the bottom 64 kilobytes: address \$FFFFFF00 reads the same byte as \$0000FF00.

This alias exists for the 68K. A pre-decrement from a register holding zero (`-(SP)` with `SP = 0`) wraps to \$FFFFFFFC, which must still hit the stack. The alias makes that work.

## 24.2 Main RAM

Main RAM begins at \$00000000 and runs up to the bottom of the I/O region.

|            |                |                                                   |
|------------|----------------|---------------------------------------------------|
| \$00000000 | Vector table   | (interrupt vectors and reserved slots)            |
| \$00001000 | Program code   | ( <code>PROG_START</code> )                       |
| \$00002000 | Stack region   | (grows downward from <code>STACK_START</code> )   |
| \$0009F000 | Top of stack   | ( <code>STACK_START</code> - initial SP for IE32) |
| \$0009FFFF | End of low RAM |                                                   |

The vector table layout depends on which CPU is active. The 68K follows the standard 68000 vector table (Chapter 29). The 6502 keeps its NMI/RESET/IRQ vectors in the top six bytes of the 16-bit page that maps to \$0000FFFA-\$0000FFFF. IE64 and IE32 use their own vector formats (Chapters 25 and 26).

`PROG_START` at \$1000 is the conventional low-RAM entry point for monitor-entered and loaded machine-code programs. Programs are free to extend past \$1000 upward and to use the stack region below it for data, but the floor at \$2000 is the conventional ceiling for the stack and the floor for static data. Programs that need more stack than 637 kilobytes can simply place SP higher in extended RAM.

## 24.3 The MMIO region

Everything between \$F0000 and \$FFFF, together with the two VGA legacy VRAM apertures at \$A0000 and \$B8000 and the ULA VRAM aperture at \$FA000, is memory-mapped I/O: the registers of the six picture chips, the ten audio engines,

the file system bridge, the coprocessor, and various small control surfaces. A read or write to an MMIO address does not touch RAM; it asks the corresponding device to do something.

The full map:

| Range           | Size  | Device                                   |
|-----------------|-------|------------------------------------------|
| \$A0000-\$AFFFF | 64K   | VGA graphics VRAM (mode \$12/\$13/\$14)  |
| \$B8000-\$BFFFF | 32K   | VGA text VRAM                            |
| \$F0000-\$F049B | 1180B | VideoChip + palette + extended blitter   |
| \$F0700-\$F07FF | 256B  | Terminal, mouse, keyboard, RTC           |
| \$F0800-\$F0B7F | 896B  | SoundChip (core six engines)             |
| \$F0BA0-\$F0BBF | 32B   | MIDI/MUS player                          |
| \$F0BC0-\$F0BD7 | 24B   | MOD player                               |
| \$F0BD8-\$F0BF3 | 28B   | WAV sample player                        |
| \$F0C00-\$F0C20 | 33B   | PSG (AY-3-8910/YM2149)                   |
| \$F0C40-\$F0CFF | 192B  | SoundChip flex channels 4-6              |
| \$F0D00-\$F0D20 | 33B   | POKEY                                    |
| \$F0D40-\$F0DFF | 192B  | SoundChip flex channels 7-9              |
| \$F0E00-\$F0E2D | 46B   | SID (6581/8580)                          |
| \$F0E80-\$F0EFF | 128B  | SFX trigger legacy aliases, channels 0-3 |
| \$D0000-\$DFFFF | 64K   | Voodoo texture memory                    |
| \$F0F00-\$F0F1F | 32B   | TED audio                                |
| \$F0F20-\$F0F6B | 76B   | TED video                                |
| \$F1000-\$F13FF | 1K    | VGA registers                            |
| \$F1400-\$F140F | 16B   | System helper MMIO                       |
| \$F2000-\$F2017 | 24B   | ULA registers                            |
| \$F2100-\$F213F | 64B   | ANTIC                                    |
| \$F2140-\$F21FB | 188B  | GTIA                                     |
| \$F2200-\$F221F | 32B   | File I/O                                 |
| \$F2260-\$F22AF | 80B   | Paula DMA (audio)                        |
| \$F2300-\$F231F | 32B   | Media loader                             |
| \$F2320-\$F233F | 32B   | RUN loader block                         |
| \$F2340-\$F238F | 80B   | Coprocessor control                      |
| \$F2390-\$F23AF | 32B   | Clipboard bridge                         |
| \$F23B0-\$F23BF | 16B   | Coprocessor extended monitor             |
| \$F23C0-\$F23DF | 32B   | IRQ diagnostics                          |
| \$F23E0-\$F23FF | 32B   | Bootstrap file bridge                    |

| Range           | Size  | Device                                     |
|-----------------|-------|--------------------------------------------|
| \$F2400-\$F24FF | 256B  | SysInfo (RAM-size discovery)               |
| \$F2600-\$F29FF | 1K    | SFX trigger extended window, channels 0-31 |
| \$F8000-\$F87FF | 2K    | Voodoo 3D registers                        |
| \$FA000-\$FBFFF | 6912B | ULA VRAM (bitmap + attrs)                  |

A few small ranges between these blocks read as zero on read and ignore on write. They are reserved for future devices.

### 24.3.1 Width and alignment

MMIO width depends on the device.

| Width   | BASIC form                                                | Typical use                                                                           |
|---------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|
| 8 bits  | POKE8 / PEEK8                                             | SID, TED audio, POKEY, PSG, SN76489 ports, palette bytes, character memory.           |
| 16 bits | POKE16 / PEEK16                                           | Some CPU-friendly aliases and split register writes.                                  |
| 32 bits | POKE32 / PEEK32                                           | Pointers, lengths, control words, status words, framebuffer bases, blitter registers. |
| 64 bits | POKE64 / PEEK64 or CPU doubleword stores where documented | Native 64-bit memory blocks and split high/low register pairs.                        |

Do not widen a byte register just because the address is in the MMIO region. POKEY, TED audio, SID, PSG, and many VGA registers are byte-oriented. Use the owning chapter's table.

PEEK16/POKE16, PEEK32/POKE32, and PEEK64/POKE64 require 2-, 4-, and 8-byte aligned addresses. PEEK, POKE, PEEK8, and POKE8 are byte-width forms and accept any byte address.

Multi-byte player registers use little-endian byte order when you write their individual bytes. For example, a pointer value \$00100000 staged through byte writes goes low byte first:

```

10 REM STAGE MOD POINTER $00100000
20 POKE8 &H000F0BC0,0
30 POKE8 &H000F0BC1,0
40 POKE8 &H000F0BC2,16
50 POKE8 &H000F0BC3,0
60 PRINT PEEK8(&H000F0BC0),PEEK8(&H000F0BC1)
70 PRINT PEEK8(&H000F0BC2),PEEK8(&H000F0BC3)

```

A 32-bit POKE32 &H000F0BC0,&H00100000 writes the same staged pointer in one operation.

Lines 20 to 50 write the four bytes of the pointer from least significant byte to most significant byte. Lines 60 and 70 read those bytes back in two print zones, so the expected values are 0, 0, then 16, 0.

64-bit accesses to a legacy MMIO region are not the normal reader workflow. Use the documented 32-bit high/low registers, such as SYSINFO\_TOTAL\_RAM\_LO and SYSINFO\_TOTAL\_RAM\_HI, unless a chip chapter explicitly documents a 64-bit register.

## 24.4 The terminal region

\$F0700-\$F07FF holds the basic human-facing peripherals: the text terminal, mouse, keyboard scancodes, and a real-time clock.

| Address | Name             | Description                                               |
|---------|------------------|-----------------------------------------------------------|
| \$F0700 | TERM_OUT         | Write a byte: emit to terminal                            |
| \$F0704 | TERM_STATUS      | b0 input ready, b1 output ready                           |
| \$F0708 | TERM_IN          | Read next input byte (dequeues)                           |
| \$F070C | TERM_LINE_STATUS | b0 complete line available                                |
| \$F0710 | TERM_ECHO        | b0 local echo enable (default 1)                          |
| \$F0724 | TERM_CTRL        | b0 line input mode                                        |
| \$F0728 | TERM_KEY_IN      | Read next raw key byte                                    |
| \$F072C | TERM_KEY_STATUS  | b0 raw key available                                      |
| \$F0730 | MOUSE_X          | Absolute X position                                       |
| \$F0734 | MOUSE_Y          | Absolute Y position                                       |
| \$F0738 | MOUSE_BUTTONS    | b0 left, b1 right, b2 middle                              |
| \$F073C | MOUSE_STATUS     | b0 changed-since-last-read                                |
| \$F074C | MOUSE_CTRL       | b0 request relative/captured mode                         |
| \$F0740 | SCAN_CODE        | Dequeue raw keyboard scancode                             |
| \$F0744 | SCAN_STATUS      | b0 scancode available                                     |
| \$F0748 | SCAN_MODIFIERS   | b0 shift, b1 ctrl, b2 alt, b3 caps                        |
| \$F0750 | RTC_EPOCH        | UTC seconds since 1970-01-01                              |
| \$F0754 | MOUSE_DX         | Signed relative X delta (clears)                          |
| \$F0758 | MOUSE_DY         | Signed relative Y delta (clears)                          |
| \$F075C | RTC_MONO_USEC_LO | Low 32 bits of monotonic microseconds since engine start  |
| \$F0760 | RTC_MONO_USEC_HI | High 32 bits of monotonic microseconds since engine start |
| \$F07F0 | TERM_SENTINEL    | Write \$DEAD to halt the CPU                              |

The M68K reaches TERM\_OUT through a second alias as well:

- TERM\_OUT\_SIGNEXT = \$FFFF700: the sign-extended .W form that an M68K MOVE.B D0, (\$F700).W resolves to. The bus folds this back onto the same \$F0700 register through the sign-extended mirror described in section 24.1.2.

The 6502 and Z80 have no equivalent 16-bit alias for TERM\_OUT: the only 16-bit address that would translate to \$F0700 is \$F700, which the CPU adapters intercept as BANK1\_REG\_L0 before the bus translation runs. Code that needs the system terminal can select TERM\_IO\_BANK into bank 1 with SET\_TERMINAL\_BANK, then use the 6502/Z80 include-file terminal aliases at \$2700-\$27FF; TERM\_OUT is \$2700 in that window and resolves to bus \$F0700. See Chapters 27 and 28 for the per-CPU details.

Chapter 37 covers the keyboard, mouse, and controller MMIO in detail. Chapter 38 covers the serial interface that overlays TERM\_OUT/TERM\_IN.

## 24.5 The system-information block

\$F2400-\$F24FF. Six read-only words let a program discover how much memory it has to play with:

| Address | Name                  | Description                                  |
|---------|-----------------------|----------------------------------------------|
| \$F2400 | SYSINFO_TOTAL_RAM_LO  | Low 32 bits of total RAM, bytes              |
| \$F2404 | SYSINFO_TOTAL_RAM_HI  | High 32 bits of total RAM                    |
| \$F2408 | SYSINFO_ACTIVE_RAM_LO | Low 32 bits of RAM visible to the active CPU |
| \$F240C | SYSINFO_ACTIVE_RAM_HI | High 32 bits of CPU-visible RAM              |
| \$F2414 | SYSINFO_LOW_WINDOW_LO | Low 32 bits of the dense low RAM window      |
| \$F2418 | SYSINFO_LOW_WINDOW_HI | High 32 bits of the dense low RAM window     |

The total and active values can differ when a 16-bit profile (6502 or Z80) is the active CPU: total reports the physical RAM, active reports the window the small CPU can currently see. The low-window value is different again. It tells software how much contiguous low RAM is backed by the ordinary low address path. Use it when a buffer must be reachable by a 32-bit image, a compatibility CPU, or a device register that stores a low-window pointer.

Type this to print both low words. On machines with more than 4 GB, the high words at lines 30 and 50 are non-zero.

```
10 REM SYSINFO RAM SIZE WORDS
20 TL=PEEK32(&H000F2400)
30 TH=PEEK32(&H000F2404)
40 AL=PEEK32(&H000F2408)
50 AH=PEEK32(&H000F240C)
60 PRINT "TOTAL LO/HI ";TL,TH
70 PRINT "ACTIVE LO/HI ";AL,AH
```

Writes to the SysInfo block are ignored. It is a discovery block, not a configuration block.

The low word is enough on ordinary small configurations. The high word is there so the same code can describe a larger memory configuration without changing the ABI.

## 24.6 The bootstrap file bridge

\$F23E0-\$F23FF. This eight-register block exposes the file system to bootstrap code that runs before BASIC is up.

| Offset | Name | Direction | Description              |
|--------|------|-----------|--------------------------|
| +0     | CMD  | W         | Command code, 0-7        |
| +4     | ARG1 | W         | First argument           |
| +8     | ARG2 | W         | Second argument          |
| +12    | ARG3 | W         | Third argument           |
| +16    | ARG4 | W         | Fourth argument          |
| +20    | RES1 | R         | First result             |
| +24    | RES2 | R         | Second result            |
| +28    | ERR  | R         | Error code (0 = success) |

Command codes are 0 DISCOVER, 1 OPEN, 2 READ, 3 CLOSE, 4 STAT, 5 READDIR, 6 CREATE\_WRITE, 7 WRITE. Chapter 35 has the full protocol.

Most programs never touch this region directly. BASIC's LOAD, SAVE, BLOAD, COMPILE, TRANSPILE, ASSEMBLE, DIR, and TYPE go through it, as does the program loader. It is here for the rare case where you are writing your own loader.

The File I/O block carries its data address through a 32-bit register. A read, write, or directory listing is refused if the staged buffer would cross the sign-extended alias guard at \$FFFF0000 or run past active RAM. In that case the file block reports FILE\_ERR\_RANGE instead of wrapping into low memory or copying only part of the transfer. Chapter 35 gives the register-level details.

## 24.7 The IRQ diagnostics block

\$F23C0-\$F23DF. Eight read-only words that report interrupt activity. These are diagnostic taps; ordinary programs do not need them.

| Address | Name                | Reports                                            |
|---------|---------------------|----------------------------------------------------|
| \$F23C0 | IRQ_DIAG_ISR        | Interrupt-in-service bitmask                       |
| \$F23C4 | IRQ_DIAG_FLAGS      | b0 stopped, b1 in-exception, b2 INTENA, b3 running |
| \$F23C8 | IRQ_DIAG_PENDING    | Pending interrupt bitmask                          |
| \$F23CC | IRQ_DIAG_COUNTERS   | L5 delivered (lo16) + L4 delivered (hi16)          |
| \$F23D0 | IRQ_DIAG_BLOCKED    | L5 blocked (lo16) + L4 blocked (hi16)              |
| \$F23D4 | IRQ_DIAG_RTE        | RTE count                                          |
| \$F23D8 | IRQ_DIAG_STOP_SPINS | Consecutive STOP iterations                        |
| \$F23DC | IRQ_DIAG_WATCHDOG   | Latched watchdog event count                       |

Chapter 31 describes the trap and exception model that backs these counters.

## 24.8 PEEK and POKE from BASIC

BASIC has byte-width historical primitives and explicit wider forms:

- PEEK(addr) and PEEK8(addr) read an 8-bit unsigned value. POKE addr, value and POKE8 addr, value write one byte. These accept any address and are the right choice for byte-oriented MMIO registers such as TED audio (\$F0F00-\$F0F05), the WAV volume bytes (\$F0BF1-\$F0BF2), the SID register file, and any single byte of the VGA palette.
- PEEK32(addr) reads a 32-bit unsigned value. POKE32 addr, value writes a 32-bit unsigned value. Both require addr to be a multiple of 4; an unaligned address raises ?FC ERROR.
- PEEK64(addr) reads a 64-bit value from an 8-byte aligned address. It returns an exact integer qword, so HEX\$(PEEK64(addr)), POKE64 dst, PEEK64(src), and other integer-compatible expressions preserve pointer and bitfield payloads.

```
10 REM READ THE VBLANK FLAG FROM VIDEOCHIP
20 V=PEEK32(&H000F0008)
30 IF (V AND 1)=0 THEN GOTO 20
40 REM DO SOMETHING AT THE START OF VBLANK
```

This is a 32-bit status register, so PEEK32 is the right access width. The loop waits until bit 0 becomes non-zero.

```
10 REM EMIT A BANNER ONE BYTE AT A TIME
20 B$="INTUITION ENGINE"
30 FOR I=1 TO LEN(B$)
40 POKE8 &H000F0700,ASC(MID$(B$,I,1))
50 NEXT I
60 POKE8 &H000F0700,13
```

TERM\_OUT is a byte output register. The loop writes one character code at a time, then line 60 writes carriage return. Use POKE8 here; a 32-bit POKE would still deliver a low byte, but it hides the fact that this is a byte device.

The MMIO map decides what a read actually returns. \$F0700 is the TERM\_OUT register: it is write-only and reads back zero. To poll terminal status, read TERM\_STATUS at \$F0704 (bit 0 input available, bit 1 output ready). Tables in the following chapters mark which MMIO addresses are write-only and which return a stable value on read.

## 24.9 Addresses on the smaller CPUs

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The 6502 and Z80 cannot see more than 64 kilobytes at a time. For them, the I/O region appears at the top of the 16-bit space:

- 6502: addresses \$F000-\$FFF9 map to \$F0000-\$F0FF9, except for the bank-register page \$F700-\$F705 and the VRAM\_BANK\_REG at \$F7F0, which the CPU adapter intercepts. \$FFFA-\$FFFF is the 6502 vector table and is not translated. The VGA is reached at \$D700-\$D70D; the ULA at \$D800-\$D817 with a paged VRAM port. Chapter 27 lists the full set of aliases.
- Z80: same \$F000-\$FFF9 MMIO window with the same \$F700 bank-register intercept. The Z80 also exposes the VGA through OUT (\$A0..\$AA) port I/O, and the ULA through ports \$FE/\$FD/\$BE/\$FA-\$FC with a paged VRAM port.

IE32, M68K, and x86 see the low 32-bit bus window directly. IE64 sees that same low window and can also reach RAM above it through its 64-bit physical path. x86 data accesses may use the native MMIO addresses at \$000F0000-\$000FFFFFF; the \$F000-\$FFFF compatibility mirror is a data-access mirror only. Instruction fetch at \$F000 reads program RAM at \$0000F000. A POKE at \$000F0700 from BASIC, a 68K MOVE.L D0, (\$F0700).L, and an x86 data store to \$000F0700 reach the same terminal register because all three names land inside the common low MMIO window.

## 24.10 What comes next

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Chapter 25 begins the per-CPU section with IE64, the native processor of the Intuition Engine. It is the easiest CPU to write for because it has the most general instruction set and the most registers; everything else in Part IV is the same story told through a smaller instruction set.